Julien Soulé

Doctor in Computer Science
AI & Multi-Agent Systems

2 Boulevard Robert Schuman
57100 Thionville, France
+33 (0) 6 77 63 12 13
julien.soule@hotmail.fr
julien6.github.io/home/
julien-soulé-6b2b27173
homepages/367/9947
0009-0002-3218-2614
julien6

[Updated on July 9, 2026]

Research Profile

I am a Doctor in Computer Science specializing in Multi-Agent Systems and AI with a focus on Cyberdefense. My research lies at the intersection of neuro-symbolic AI, multi-agent reinforcement learning, and world models, combining symbolic reasoning with machine learning to develop intelligent, explainable, and resilient autonomous systems. My goal is to enable the design, deployment, and lifelong adaptation of trustworthy multi-agent systems through human-in-the-loop supervision and formal guarantees across their entire lifecycle.

Research Interests

Multi-Agent Reinforcement Learning (MARL); Multi-Agent System Organizations; Neuro-Symbolic AI; Cyberdefense & Intelligent Cyber Agents; World Models & Digital Twins; Software Architecture for MAS.

Education

2022–2025 **PhD in Computer Science**, *Université Grenoble Alpes (UGA)*, France
Dissertation defended on **17 November 2025**. Title: *"On the Organization of a Cyberdefence Multi-Agent System"*. Status: **Degree conferred**. Supervisors: Jean-Paul Jamont, Michel Occello, Louis-Marie Traonouez, Paul Théron.

2018–2021 **Master's Degree in Computer Engineering**, *INSA Rennes*, France
Specialization in Computer Science and Cybersecurity.

2019 **Exchange Semester in Computer Engineering**, *École de Technologie Supérieure (ETS)*, Montréal, Canada

2015–2018 **Integrated Preparatory Cycle**, *INSA Rennes*, France

2012–2015 **High School**, *Lycée Charles Renouvier*, Prades, France

Appointments

2026–present **Postdoctoral Researcher**, *University of Luxembourg*, Luxembourg
Conduct research on neuro-symbolic approaches for Multi-Agent Systems, MARL, World Models and application for industrial projects, especially with *Thales Cyber Solutions* and the Luxembourgish police.

2022–2025 **Doctoral Researcher**, *Thales Land & Air Systems + Université Grenoble Alpes*, France
Conducted doctoral research on organizational multi-agent systems, MARL, and cyberdefense architectures. PhD defended on 17 November 2025.

2021–2022 **Research Engineer**, *Thales Land & Air Systems*, Rennes, France
Study on multi-agent modeling of cyber environments and anomaly detection in Thales LAS context.

2020–2021 **Software and Research Engineer**, *Atos*, Toulouse, France
Software engineering for airspace systems developed with CNES and ESA.

2020 **Software Engineering Internship**, *Atos*, Toulouse, France
Development of the ISIS system for satellite command & control (Python, Bash, KVM, Grafana, Django).

2019 **Software Engineering Internship**, *SQLI*, Toulouse, France
Cybersecurity, maintenance, and software engineering tasks for Airbus Helicopters projects.

Research Projects

ALIAS A research project developed within *Thales Cyber Solutions* to support operators in External Attack Surface Management (EASM). The platform combines reinforcement learning, multi-agent reinforcement learning, world models, large language models, and neuro-symbolic AI within a human-in-the-loop framework to enable reliable, explainable, and adaptive cyber defense.

AssistDiFo A research project conducted in collaboration with the Luxembourg Police to assist investigators in analyzing large volumes of heterogeneous digital evidence. The platform leverages named entity recognition, retrieval-augmented generation (RAG), and large language models while maintaining human oversight throughout the investigation process.

CybMASDE A research platform combining MARL, organizational modeling, and cyberdefense simulation to support the design and analysis of intelligent multi-agent architectures.

MOISE+MARL A proof-of-concept framework integrating organizational roles, missions, and goals into MARL environments to guide learning and improve agent coordination.

Publications

Journal Articles

[2] Julien Soulé et al. "Assisting Multi-Agent System Design with $\mathcal{MOISE}^{+}$ and MARL: The MAMAD Method". In: *Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS)* (2026). URL: `https://link.springer.com/article/10.1007/s10458-026-09740-0`.

[4] Julien Soulé et al. "MOISE+MARL : un cadre organisationnel pour l'explicabilité et le contrôle en apprentissage par renforcement multi-agent". In: *Revue Ouverte d'Intelligence Artificielle (ROIA)* (2025). *Accepted for publication*.

International Conferences and Workshops

[1] Julien Soulé. "Toward Automated Operational Assist in Satellite Fleet Management via Organizational MARL". In: *Proceedings of the International Workshop on Autonomous Agents and Multi-Agent Systems for Space Applications (**MASSpace-26**)*. Held as part of the Workshops at the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026), May 26, 2026. Paphos, Cyprus: International Foundation for Autonomous Agents and Multiagent Systems, May 2026. URL: `https://masspace.github.io/aamas2026ws/papers/MASSpace-2026-Proceedings.pdf#page=96`.

[3] Julien Soulé et al. "An Organizationally-Oriented Approach to Enhancing Explainability and Control in Multi-Agent Reinforcement Learning". In: *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS** 2025)*. Detroit, MI, USA: International Foundation for Autonomous Agents and Multiagent Systems, 2025, pp. 1968–1976. ISBN: 9798400714269.

[5] Julien Soulé et al. "Streamlining Resilient Kubernetes Autoscaling with Multi-Agent Systems via an Automated Online Design Framework". In: *2025 IEEE 18th International Conference on Cloud Computing (**CLOUD** 2025)*. 2025, pp. 43–53. DOI: `10.1109/CLOUD67622.2025.00015`.

[7] Julien Soulé et al. "A MARL-Based Approach for Easing MAS Organization Engineering". In: *Artificial Intelligence Applications and Innovations (**AIAI** 2024)*. Ed. by Ilias Maglogiannis et al. Cham: Springer Nature Switzerland, 2024, pp. 321–334. ISBN: 978-3-031-63223-5.

[14] Julien Soulé et al. "Towards a Multi-Agent Simulation of Cyber-attackers and Cyber-defenders Battles". In: *2023 IEEE International Conference on Systems, Man, and Cybernetics (**SMC** 2023)*. 2023, pp. 3594–3599. DOI: `10.1109/SMC53992.2023.10394564`.

National Conferences (France)

[6] Julien Soulé et al. “Une approche organisationnelle pour améliorer l’explicabilité et le contrôle dans l’apprentissage par renforcement multi-agent”. In: *33èmes Journées Francophones sur les Systèmes Multi-Agents (**JFSMA** 2025)*. **Best paper award**. Dijon, France: Association Française pour l’Intelligence Artificielle, July 2025. URL: `https://hal.science/hal-05151654`.

[8] Julien Soulé et al. “Un outil pour la conception de SMA par apprentissage par renforcement et modélisation organisationnelle”. In: *32èmes Journées Francophones sur les Systèmes Multi-Agents (**JFSMA** 2024)*. Sébastien Picault. Cargèse, France: Cépaduès, Nov. 2024. URL: `https://hal.science/hal-04840721`.

[9] Julien Soulé et al. “Une approche basée sur l’apprentissage par renforcement pour l’ingénierie organisationnelle d’un SMA”. In: *32èmes Journées Francophones sur les Systèmes Multi-Agents (**JFSMA** 2024)*. Sébastien Picault. Cargèse, France: Cépaduès, Nov. 2024. URL: `https://hal.science/hal-04840696`.

[10] Julien Soulé et al. “De l’organisation d’un système multi-agent de cyberdéfense”. In: *31èmes Journées Francophones sur les Systèmes Multi-Agents (**JFSMA** 2023)*. JFSMA. Cépaduès, 2023, pp. 54–54. URL: `https://hal.science/hal-04165020`.

[11] Julien Soulé et al. “De l’organisation d’un système multi-agent de cyberdéfense”. In: *Rendez-Vous de la Recherche et de l’Enseignement de la Sécurité des Systèmes d’Information (**RESSI** 2023)*. May 2023. URL: `https://ressi2023.sciencesconf.org/450961/document`.

[12] Julien Soulé et al. “De l’organisation des systèmes multi-agents de cyber-défense”. In: *Rencontres Jeunes Chercheurs en Intelligence Artificielle (**RJCIA** 2023)*. Strasbourg, France, July 2023. URL: `https://hal.science/hal-04565426`.

Invited Talks

[13] Julien Soulé et al. *Talk for the **CybAIR NATO Chair***. École de l’air et de l’espace. Mar. 2023.

Teaching Experience

2023–2025 **Teaching Assistant**, *Valence (UGA Engineering School & IUT)*
Tutorials and labs in Operating Systems, Systems Programming, and Process Management. Supervision of student groups working on industry-driven cyberdefense projects.

Service

2023–Present Treasurer of the Autonomous Intelligent Cyberdefence Agent (AICA) International Work Group.

Reviewing Activities

2026 Reviewer for the 22nd International Conference on Network and Service Management (**CNSM** 2026).

2025 Reviewer for the 17th International Conference on Agents and Artificial Intelligence (**ICAART** 2025).

2025 Reviewer for the 25th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS** 2026).

2025 2025 IEEE International Conference on Development and Learning (**ICDL** 2025).

2024 Reviewer for the 16th International Conference on Agents and Artificial Intelligence (**ICAART** 2024).

2024 Reviewer for the 2025 IEEE International Conference on Systems, Man, and Cybernetics (**SMC** 2025).

Awards and Distinctions

2025 Best Paper Award at JFSMA 2025 for *“Une approche organisationnelle pour améliorer l’explicabilité et le contrôle dans l’apprentissage par renforcement multi-agent”*.

Skills

Languages French (native), English (professional, TOEIC 910), Spanish (intermediate),, Japanese (basic).